

App Server and Web Server Task

1.Install nginx and run nginx on port number 81.

```
root@ip-172-31-22-21:~
Last metadata expiration check: 2:09:44 ago on Tue Jun 30 06:42:11 2026.
Dependencies resolved.
=====
Package                Architecture Version                Repository            Size
=====
Installing:
nginx                  x86_64         1:1.30.2-1.amzn2023.0.1  amazonlinux           34 k
Installing dependencies:
gperftools-libs        x86_64         2.9.1-1.amzn2023.0.3    amazonlinux           308 k
libunwind              x86_64         1.4.0-5.amzn2023.0.3    amazonlinux           66 k
nginx-core             x86_64         1:1.30.2-1.amzn2023.0.1  amazonlinux           709 k
nginx-filesystem       noarch         1:1.30.2-1.amzn2023.0.1  amazonlinux           10 k
nginx-mimetypes        noarch         2.1.49-3.amzn2023.0.3    amazonlinux           21 k
=====
Transaction Summary
=====
Install 6 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Is this ok [y/N]: y
Downloading Packages:
(1/6): nginx-1.30.2-1.amzn2023.0.1.x86_64.rpm           864 kB/s | 34 kB      00:00
(2/6): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 6.5 MB/s | 308 kB    00:00
(3/6): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm       1.3 MB/s | 66 kB     00:00
(4/6): nginx-core-1.30.2-1.amzn2023.0.1.x86_64.rpm     21 MB/s | 709 kB     00:00
(5/6): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm 790 kB/s | 21 kB     00:00
(6/6): nginx-filesystem-1.30.2-1.amzn2023.0.1.noarch.rpm 354 kB/s | 10 kB     00:00
-----
Total                                           9.5 MB/s | 1.1 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                : 1/1
  Running scriptlet: nginx-filesystem-1:1.30.2-1.amzn2023.0.1.noarch 1/6
  Installing          : nginx-filesystem-1:1.30.2-1.amzn2023.0.1.noarch 1/6
  Installing          : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch 2/6
  Installing          : libunwind-1.4.0-5.amzn2023.0.3.x86_64 3/6
  Installing          : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 4/6
  Installing          : nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64 5/6
  Installing          : nginx-1:1.30.2-1.amzn2023.0.1.x86_64 6/6
  Running scriptlet: nginx-1:1.30.2-1.amzn2023.0.1.x86_64 6/6
  Verifying           : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 1/6
  Verifying           : libunwind-1.4.0-5.amzn2023.0.3.x86_64 2/6
  Verifying           : nginx-1:1.30.2-1.amzn2023.0.1.x86_64 3/6
  Verifying           : nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64 4/6
  Verifying           : nginx-filesystem-1:1.30.2-1.amzn2023.0.1.noarch 5/6
  Verifying           : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch 6/6
Installed:
gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 libunwind-1.4.0-5.amzn2023.0.3.x86_64
nginx-1:1.30.2-1.amzn2023.0.1.x86_64 nginx-core-1:1.30.2-1.amzn2023.0.1.x86_64
nginx-filesystem-1:1.30.2-1.amzn2023.0.1.noarch nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[root@ip-172-31-22-21 ~]#
```

Not secure 18.206.174.8

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.

To engage with the community please visit community.nginx.org.

For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

```
root@ip-172-31-22-21:/usr/share/nginx/html
[root@ip-172-31-22-21 ~]# cd /usr/share/nginx/html
bash: cd: /usr/share/nginx/html: No such file or directory
[root@ip-172-31-22-21 ~]# cd /usr/share/nginx/html
[root@ip-172-31-22-21 html]# ls
404.html 50x.html icons index.html nginx-logo.png poweredby.png
[root@ip-172-31-22-21 html]# cat index.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
[root@ip-172-31-22-21 html]#
```

Not secure 18.206.174.8

Welcome to Techie Horizon!!!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

```
[root@ip-172-31-22-21 ~]# cd /etc/nginx
[root@ip-172-31-22-21 nginx]# ls
conf.d          fastcgi_params  mime.types      scgi_params     win-utf
default.d       fastcgi_params.default  mime.types.default  scgi_params.default  win-utf
fastcgi.conf    koi-utf        nginx.conf      uwsgi_params
fastcgi.conf.default  koi-win       nginx.conf.default  uwsgi_params.default
[root@ip-172-31-22-21 nginx]# cp nginx.conf /opt/
[root@ip-172-31-22-21 nginx]# cd /opt
[root@ip-172-31-22-21 opt]# ls
aws  nginx.conf
[root@ip-172-31-22-21 opt]# cd /etc/nginx
[root@ip-172-31-22-21 nginx]# mv nginx.conf nginx.conf.backup
[root@ip-172-31-22-21 nginx]# ls
conf.d          fastcgi_params  mime.types      scgi_params     win-utf
default.d       fastcgi_params.default  mime.types.default  scgi_params.default  win-utf
fastcgi.conf    koi-utf        nginx.conf.backup  uwsgi_params
fastcgi.conf.default  koi-win       nginx.conf.default  uwsgi_params.default
[root@ip-172-31-22-21 nginx]#
```

```

root@ip-172-31-22-21:/etc/nginx
types_hash_max_size 4096;

include /etc/nginx/mime.types;
default_type application/octet-stream;

# Load modular configuration files from the /etc/nginx/conf.d directory.
# See http://nginx.org/en/docs/nginx_core_module.html#include
# for more information.
include /etc/nginx/conf.d/*.conf;

server {
    listen 81;
    listen [::]:81;
    server_name _;
    root /usr/share/nginx/html;

    # Load configuration files for the default server block.
    include /etc/nginx/default.d/*.conf;

    error_page 404 /404.html;
    location = /404.html {
    }

    error_page 500 502 503 504 /50x.html;
    location = /50x.html {
    }
}

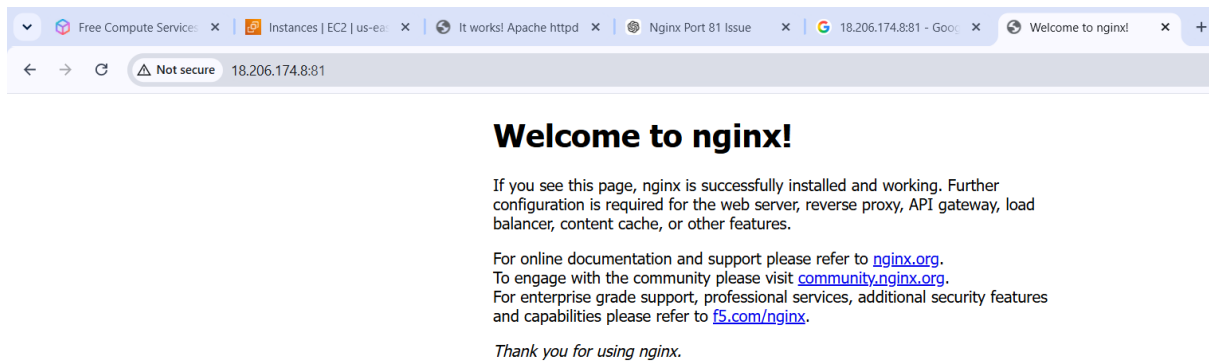
# Settings for a TLS enabled server.
#
# server {
#     listen 443 ssl;
#     listen [::]:443 ssl;
#     http2 on;
#     server_name _;
#     root /usr/share/nginx/html;
#
#     ssl_certificate "/etc/pki/nginx/server.crt";
#     ssl_certificate_key "/etc/pki/nginx/private/server.key";
#     ssl_session_cache shared:SSL:1m;
#     ssl_session_timeout 10m;
#     ssl_ciphers PROFILE=SYSTEM;
#     ssl_prefer_server_ciphers on;
#
#     # Load configuration files for the default server block.
#     include /etc/nginx/default.d/*.conf;
#
#     error_page 404 /404.html;
#     location = /404.html {
#     }
#
#     error_page 500 502 503 504 /50x.html;
#     location = /50x.html {
#     }
# }
~

```

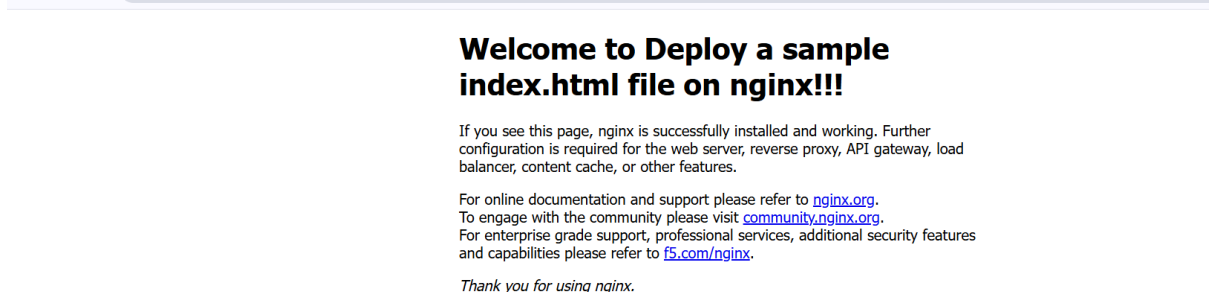
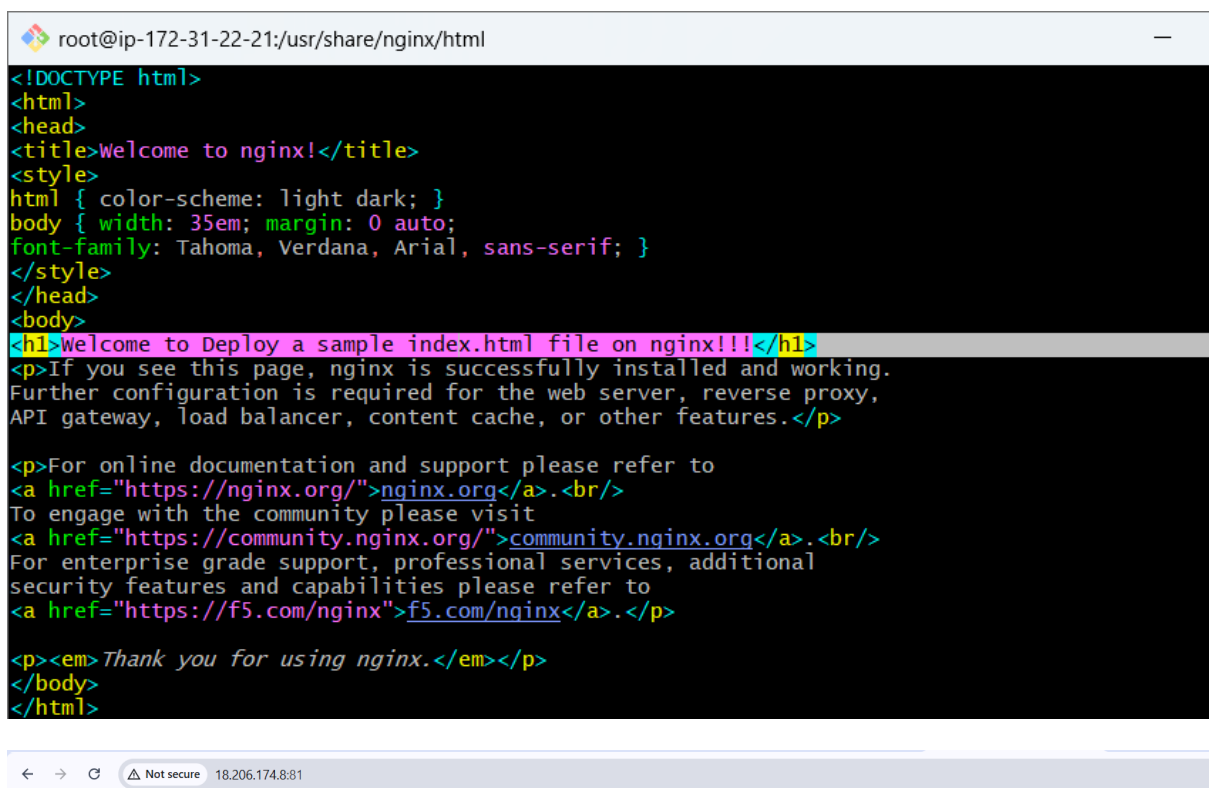
```

root@ip-172-31-22-21:/etc/nginx
[root@ip-172-31-22-21 nginx]# netstat -na | grep 80
udp6      0      0 fe80::8ff:c6ff:fe85:546 :::*
unix  2      [ ]          DGRAM          4880      /run/user/1000/systemd/notify
unix  3      [ ]          STREAM         CONNECTED   3680      /run/dbus/system_bus_socket
unix  2      [ ACC ]      STREAM         LISTENING   806       /run/systemd/private
unix  3      [ ]          STREAM         CONNECTED   36280
[root@ip-172-31-22-21 nginx]# netstat -na | grep 81
tcp        0      0 0.0.0.0:81 0.0.0.0:*      LISTEN
tcp6       0      0 :::81     :::*           LISTEN
unix  3      [ ]          STREAM         CONNECTED   98183
unix  3      [ ]          DGRAM          4881
unix  3      [ ]          STREAM         CONNECTED   98184
unix  2      [ ]          DGRAM          98178
[root@ip-172-31-22-21 nginx]#

```



2. Deploy a sample index.html file on nginx.



3. Install Apache and run Apache on port number 82

```
root@ip-172-31-28-28:~  
[root@ip-172-31-28-28 ~]# sudo dnf update -y  
Last metadata expiration check: 1 day, 0:36:16 ago on Mon Jun 29 17:46:08 2026.  
=====
```

WARNING:

A newer release of "Amazon Linux" is available.

Available Versions:

Version 2023.12.20260629:
Run the following command to upgrade to 2023.12.20260629:

dnf upgrade --releasever=2023.12.20260629

Release notes:
<https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.12.20260629.html>

=====

Dependencies resolved.
Nothing to do.
Complete!

```
[root@ip-172-31-28-28 ~]# sudo dnf install httpd -y  
Last metadata expiration check: 1 day, 0:36:28 ago on Mon Jun 29 17:46:08 2026.  
Package httpd-2.4.68-1.amzn2023.0.1.x86_64 is already installed.  
Dependencies resolved.  
Nothing to do.  
Complete!
```

```
[root@ip-172-31-28-28 ~]# httpd -v  
Server version: Apache/2.4.68 (Amazon Linux)  
Server built: Jun 8 2026 00:00:00  
[root@ip-172-31-28-28 ~]# sudo systemctl start httpd  
[root@ip-172-31-28-28 ~]# sudo systemctl status httpd  
● httpd.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)  
   Active: active (running) since Fri 2026-06-26 18:39:35 UTC; 3 days ago  
     Docs: man:httpd.service(8)  
  Main PID: 26313 (httpd)  
    Status: "Total requests: 184; Idle/Busy workers 100/0; Requests/sec: 0.000534; Bytes served/sec: 1.000000"  
    Tasks: 230 (limit: 1059)  
   Memory: 19.6M  
      CPU: 6min 162ms  
   CGroup: /system.slice/httpd.service  
           └─ 26313 /usr/sbin/httpd -DFOREGROUND  
             └─ 104203 /usr/sbin/httpd -DFOREGROUND  
               └─ 104204 /usr/sbin/httpd -DFOREGROUND  
                 └─ 104232 /usr/sbin/httpd -DFOREGROUND  
                   └─ 104302 /usr/sbin/httpd -DFOREGROUND  
                     └─ 273358 /usr/sbin/httpd -DFOREGROUND
```

```
Jun 26 18:39:35 ip-172-31-28-28.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Serv  
Jun 26 18:39:35 ip-172-31-28-28.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Serve  
Jun 26 18:39:35 ip-172-31-28-28.ec2.internal httpd[26313]: Server configured, listening on: port 80  
Jun 28 00:00:05 ip-172-31-28-28.ec2.internal systemd[1]: Reloading httpd.service - The Apache HTTP Ser  
Jun 28 00:00:05 ip-172-31-28-28.ec2.internal systemd[1]: Reloaded httpd.service - The Apache HTTP Serv  
Jun 28 00:00:06 ip-172-31-28-28.ec2.internal httpd[26313]: Server configured, listening on: port 80  
lines 1-23/23 (END)
```

```
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts. See the httpd.service(8) man
# page for more information.
#
#Listen 12.34.56.78:80
Listen 82

#
# Dynamic Shared Object (DSO) Support
#
# To be able to use the functionality of a module which was built as a DSO you
# have to place corresponding 'LoadModule' lines at this location so the
# directives contained in it are actually available _before_ they are used.
# Statically compiled modules (those listed by 'httpd -l') do not need
# to be loaded here.
#
# Example:
# LoadModule foo_module modules/mod_foo.so
#
Include conf.modules.d/*.conf
```

```
root@ip-172-31-28-28:~
[root@ip-172-31-28-28 ~]# sudo vi /etc/httpd/conf/httpd.conf
[root@ip-172-31-28-28 ~]# sudo systemctl restart httpd
[root@ip-172-31-28-28 ~]# ss -tulnp | grep 82
tcp    LISTEN 0      511          *:82          *:.*    users:((("httpd",pid=276478,fd=4),("httpd",pid=276477,fd=4),("httpd",pid=276476,fd=4),("httpd",pid=276474,fd=4)))
[root@ip-172-31-28-28 ~]# sudo vi /etc/httpd/conf/httpd.conf
[root@ip-172-31-28-28 ~]#
```

← → ↻ ⚠ Not secure 3.87.146.30:82

It works!

4. Deploy a sample index.html file on Apache.

```
[root@ip-172-31-28-28 ~]# yum install httpd
```

```
[root@ip-172-31-28-28 ~]# cd /var/www/html
```

```
[root@ip-172-31-28-28 html]# pwd
```

```
/var/www/html
```

```
[root@ip-172-31-28-28 html]# ls -l
```

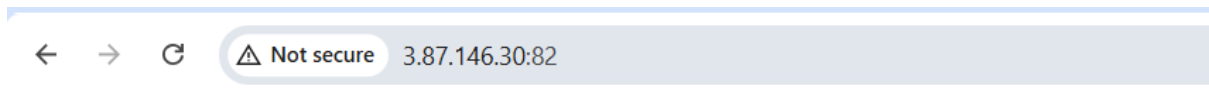
total 0

**[root@ip-172-31-28-28 html]# sudo vi
/var/www/html/index.html**

**[root@ip-172-31-28-28 html]# cat
/var/www/html/index.html**

```
root@ip-172-31-28-28:/var/www/html
[root@ip-172-31-28-28 ~]# cd /var/www/html
[root@ip-172-31-28-28 html]# pwd
/var/www/html
[root@ip-172-31-28-28 html]# ls -l
total 0
[root@ip-172-31-28-28 html]# ^C
[root@ip-172-31-28-28 html]# sudo vi /var/www/html/index.html
[root@ip-172-31-28-28 html]# cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
  <title>My Apache Server</title>
</head>
<body>
  <h1>Welcome to Apache Web Server</h1>
  <h2>Amazon Linux 2023</h2>
  <h3>Running on Port 82</h3>

  <p>Name: Badugu Rajkumari</p>
  <p>Apache Server is working successfully.</p>
</body>
</html>
[root@ip-172-31-28-28 html]# ^C
[root@ip-172-31-28-28 html]# |
```



Welcome to Apache Web Server

Amazon Linux 2023

Running on Port 82

Name: Badugu Rajkumari

Apache Server is working successfully.

5. Install Apache tomcat on port number 8082.

```
[root@ip-172-31-22-21 html]# dnf install java-17-amazon-corretto -y
Last metadata expiration check: 4:21:31 ago on Tue Jun 30 06:42:11 2026.
Dependencies resolved.
```

Package	Arch	Version	Repository	Size
Installing:				
java-17-amazon-corretto	x86_64	1:17.0.19+10-1.amzn2023.1	amazonlinux	219 k
Installing dependencies:				
alsa-lib	x86_64	1:2.7.2-1.amzn2023.0.3	amazonlinux	503 k
cairo	x86_64	1.18.0-4.amzn2023.0.3	amazonlinux	717 k
dejavu-sans-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.3 M
dejavu-sans-mono-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	467 k
dejavu-serif-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.0 M
fontconfig	x86_64	2.13.94-2.amzn2023.0.2	amazonlinux	273 k
fontsfilesystem	noarch	1:2.0.5-12.amzn2023.0.2	amazonlinux	9.5 k
freetype	x86_64	2.13.2-5.amzn2023.0.2	amazonlinux	422 k


```

Installed:
alsa-lib-1.2.7.2-1.amzn2023.0.3.x86_64
cairo-1.18.0-4.amzn2023.0.3.x86_64
dejavu-sans-fonts-2.37-16.amzn2023.0.2.noarch
dejavu-sans-mono-fonts-2.37-16.amzn2023.0.2.noarch
dejavu-serif-fonts-2.37-16.amzn2023.0.2.noarch
fontconfig-2.13.94-2.amzn2023.0.2.x86_64
fonts-filesystem-1:2.0.5-12.amzn2023.0.2.noarch
freetype-2.13.2-5.amzn2023.0.2.x86_64
giflib-5.2.1-9.amzn2023.0.4.x86_64
google-noto-fonts-common-20240401-1.amzn2023.0.2.noarch
google-noto-sans-vf-fonts-20240401-1.amzn2023.0.2.noarch
graphite2-1.3.14-7.amzn2023.0.3.x86_64
harfbuzz-7.0.0-2.amzn2023.0.2.x86_64
java-17-amazon-corretto-1:17.0.19+10-1.amzn2023.1.x86_64
java-17-amazon-corretto-headless-1:17.0.19+10-1.amzn2023.1.x86_64
javapackages-filesystem-6.0.0-7.amzn2023.0.6.noarch
langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch
libICE-1.1.1-3.amzn2023.0.1.x86_64
libSM-1.2.4-3.amzn2023.0.1.x86_64
libX11-1.8.10-2.amzn2023.0.1.x86_64
libX11-common-1.8.10-2.amzn2023.0.1.noarch
libXau-1.0.11-6.amzn2023.0.1.x86_64
libXext-1.3.6-1.amzn2023.0.1.x86_64
libXi-1.8.2-1.amzn2023.0.1.x86_64
libXinerama-1.1.5-6.amzn2023.0.1.x86_64
libXrandr-1.5.4-3.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.1.x86_64
libXt-1.3.0-3.amzn2023.0.1.x86_64
libXtst-1.2.5-1.amzn2023.0.1.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
libpng-2:1.6.37-10.amzn2023.0.13.x86_64
libxcb-1.17.0-1.amzn2023.0.1.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
[root@ip-172-31-22-21 html]# |

```

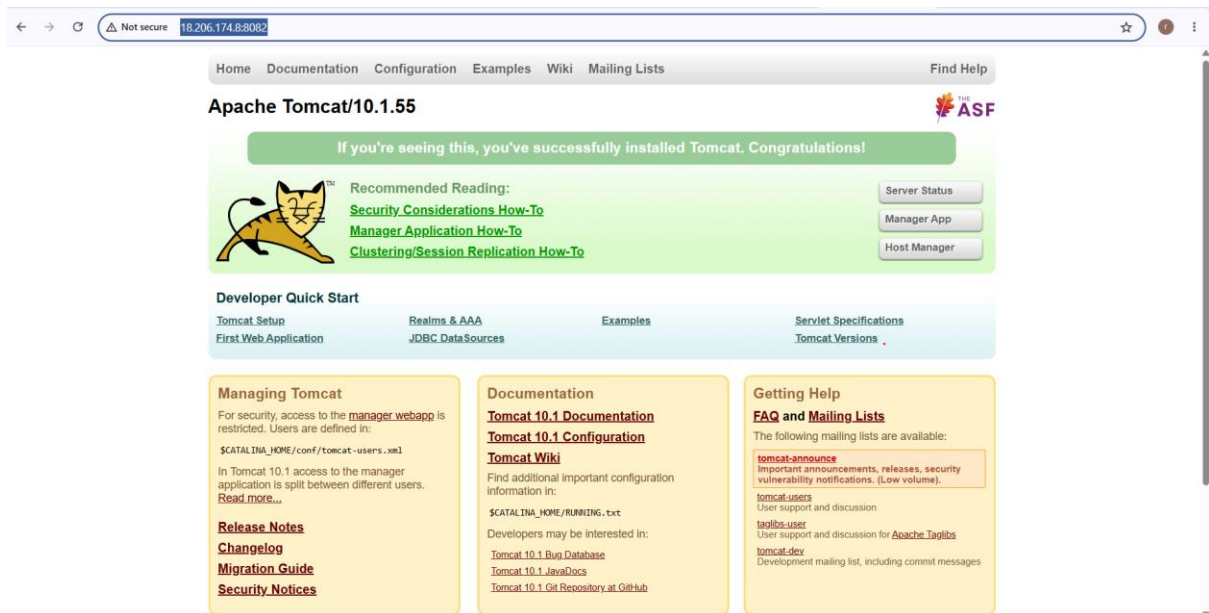
```

<!-- A "Connector" represents an endpoint by which requests are received
and responses are returned. Documentation at :
HTTP Connector: /docs/config/http.html
AJP Connector: /docs/config/ajp.html
Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
-->
<Connector port="8082" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="8443"
    maxParameterCount="1000"
/>

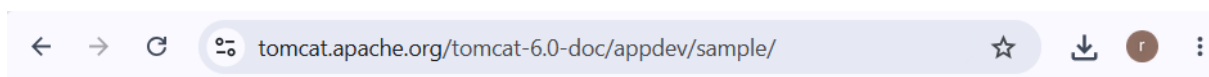
<!-- A "Connector" using the shared thread pool-->
<!--
<Connector executor="tomcatThreadPool"
    port="8080" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="8443"
    maxParameterCount="1000"
/>

-->
<!-- Define an SSL/TLS HTTP/1.1 Connector on port 8443 with HTTP/2
This connector uses the NIO implementation. The default
SSLImplementation will depend on the presence of the APR/native
library and the useOpenSSL attribute of the AprLifecycleListener.
Either JSSE or OpenSSL style configuration may be used regardless of
the SSLImplementation selected. JSSE style configuration is used below.
-->

```



6. Deploy a sample app on webapps.



Sample Application

The example app has been packaged as a war file and can be downloaded [here](#) (Note: make sure your browser doesn't change file extension or append a new one).

The easiest way to run this application is simply to move the war file to your **CATALINA_HOME/webapps** directory. Tomcat will automatically expand and deploy the application for you. You can view it with the following URL (assuming that you're running tomcat on port 8080 as is the default):
<http://localhost:8080/sample>

If you just want to browse the contents, you can unpack the war file with the **jar** command. `jar -xvf sample.war`

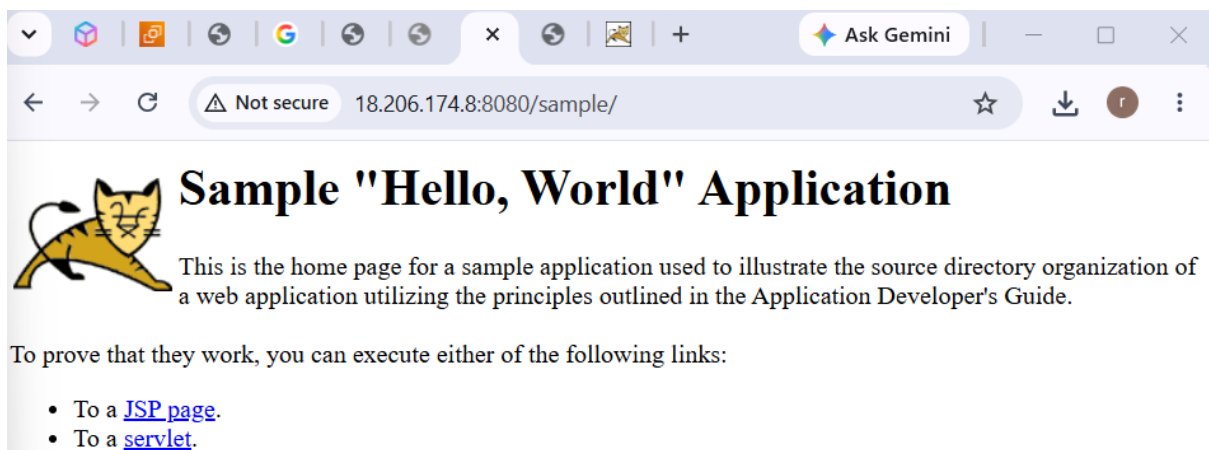
```
root@ip-172-31-22-21:/var/lib/tomcat10/webapps
[root@ip-172-31-22-21 tomcat10]# /var/lib/tomcat10/webapps
bash: /var/lib/tomcat10/webapps: Is a directory
[root@ip-172-31-22-21 tomcat10]# cd /var/lib/tomcat10/webapps
[root@ip-172-31-22-21 webapps]# ls
ROOT docs host-manager manager
[root@ip-172-31-22-21 webapps]# ll
total 32
drwxr-xr-x. 3 tomcat tomcat 16384 Jun 30 14:24 ROOT
drwxr-xr-x. 16 root root 16384 Jun 30 14:41 docs
drwxr-xr-x. 6 root tomcat 79 Jun 30 14:24 host-manager
drwxr-xr-x. 6 root tomcat 114 Jun 30 14:24 manager
[root@ip-172-31-22-21 webapps]# pwd
/var/lib/tomcat10/webapps
[root@ip-172-31-22-21 webapps]# wget https://tomcat.apache.org/tomcat-6.0-doc/appdev/sample/sample.war
--2026-06-30 14:53:42-- https://tomcat.apache.org/tomcat-6.0-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to tomcat.apache.org (tomcat.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: 'sample.war'

sample.war          100%[=====>] 4.50K  --.-KB/s  in 0s
2026-06-30 14:53:42 (54.1 MB/s) - 'sample.war' saved [4606/4606]

[root@ip-172-31-22-21 webapps]# ll
total 40
drwxr-xr-x. 3 tomcat tomcat 16384 Jun 30 14:24 ROOT
drwxr-xr-x. 16 root root 16384 Jun 30 14:41 docs
drwxr-xr-x. 6 root tomcat 79 Jun 30 14:24 host-manager
drwxr-xr-x. 6 root tomcat 114 Jun 30 14:24 manager
drwxr-xr-x. 5 tomcat tomcat 86 Jun 30 14:53 sample
-rw-r--r--. 1 root root 4606 Nov 16 2016 sample.war
[root@ip-172-31-22-21 webapps]#
```

➤ Copy the sample directory name and paste it

Example: <http://18.206.174.8:8080/sample/>





7. Create a tomcat.service file for tomcat.

```
root@ip-172-31-28-28:/opt
[root@ip-172-31-28-28 opt]# cd /opt
sudo tar -xzf apache-tomcat-10.1.42.tar.gz
[root@ip-172-31-28-28 opt]# ls
apache-tomcat-10.1.42  apache-tomcat-10.1.42.tar.gz  aws
[root@ip-172-31-28-28 opt]# sudo mv apache-tomcat-10.1.42 tomcat
[root@ip-172-31-28-28 opt]# ls /opt
apache-tomcat-10.1.42.tar.gz  aws  tomcat
[root@ip-172-31-28-28 opt]# sudo useradd -r -m -U -d /opt/tomcat -s /sbin/nologin tomcat
useradd: warning: the home directory /opt/tomcat already exists.
useradd: Not copying any file from skel directory into it.
[root@ip-172-31-28-28 opt]# sudo chown -R tomcat:tomcat /opt/tomcat
[root@ip-172-31-28-28 opt]# sudo chmod +x /opt/tomcat/bin/*.sh
[root@ip-172-31-28-28 opt]# sudo vi /etc/systemd/system/tomcat.service
[root@ip-172-31-28-28 opt]# sudo systemctl daemon-reload
[root@ip-172-31-28-28 opt]# sudo vi /etc/systemd/system/tomcat.service
[root@ip-172-31-28-28 opt]# sudo systemctl enable tomcat
Created symlink /etc/systemd/system/multi-user.target.wants/tomcat.service + /etc/systemd/system/tomcat.service.
[root@ip-172-31-28-28 opt]# sudo systemctl start tomcat
[root@ip-172-31-28-28 opt]# sudo systemctl status tomcat
● tomcat.service - Apache Tomcat Web Application Container
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-06-30 18:11:08 UTC; 8s ago
     Process: 275563 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
    Main PID: 275570 (java)
      Tasks: 29 (limit: 1059)
     Memory: 82.6M
        CPU: 4.569s
    CGroup: /system.slice/tomcat.service
            └─275570 /usr/lib/jvm/java-21-amazon-corretto.x86_64/bin/java -Djava.util.logging.config.

Jun 30 18:11:08 ip-172-31-28-28.ec2.internal systemd[1]: Starting tomcat.service - Apache Tomcat Web A
Jun 30 18:11:08 ip-172-31-28-28.ec2.internal startup.sh[275563]: Tomcat started.
Jun 30 18:11:08 ip-172-31-28-28.ec2.internal systemd[1]: Started tomcat.service - Apache Tomcat Web Ap
lines 1-14/14 (END)
```

```
root@ip-172-31-28-28:/opt
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target

[Service]
Type=forking

User=tomcat
Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/java-21-amazon-corretto.x86_64"
Environment="CATALINA_HOME=/opt/tomcat"
Environment="CATALINA_BASE=/opt/tomcat"
Environment="CATALINA_PID=/opt/tomcat/temp/tomcat.pid"

ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh

Restart=on-failure

[Install]
WantedBy=multi-user.target
~
~
~
~
```

Not secure 3.87.146.30:8080

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/10.1.42



If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

[Security Considerations How-To](#)
[Manager Application How-To](#)
[Clustering/Session Replication How-To](#)

Server Status

Manager App

Host Manager

Developer Quick Start

Tomcat Setup
First Web Application

Realms & AAA
JDBC DataSources

Examples

Servlet Specifications
Tomcat Versions

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:
`$CATALINA_HOME/conf/tomcat-users.xml`
In Tomcat 10.1 access to the manager application is split between different users.
[Read more...](#)
[Release Notes](#)
[Changelog](#)
[Migration Guide](#)
[Security Notices](#)

Documentation

[Tomcat 10.1 Documentation](#)
[Tomcat 10.1 Configuration](#)
[Tomcat Wiki](#)
Find additional important configuration information in:
`$CATALINA_HOME/RUNNING.txt`
Developers may be interested in:
[Tomcat 10.1 Bug Database](#)
[Tomcat 10.1 JavaDocs](#)
[Tomcat 10.1 Git Repository at GitHub](#)

Getting Help

[FAQ and Mailing Lists](#)
The following mailing lists are available:

[tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).

[tomcat-users](#)
User support and discussion

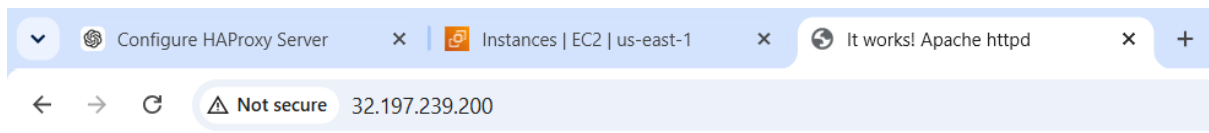
[taglibs-user](#)
User support and discussion for [Apache Taglibs](#)

[tomcat-dev](#)
Development mailing list, including commit messages

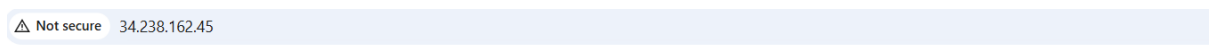
8. Configure HA Proxy server

☐	Server-2	i-00bd626b055a09aa4	 Running	 	t3.micro	 Initializing	us-east-1d	ec2-34-238-162-45.co...	3
☒	HA-Proxy-Server	i-0f0487fbe5486dbdf	 Running	 	t3.micro	 Initializing	us-east-1d	ec2-100-31-244-11.co...	1
☐	Server-1	i-0634665cf5f61922b	 Running	 	t3.micro	 Initializing	us-east-1d	ec2-32-197-239-200.co...	3

```
root@ip-172-31-35-172:~  
[root@ip-172-31-35-172 ~]# vi /etc/hosts  
[root@ip-172-31-35-172 ~]# ping load_balancer -c4  
ping: load_balancer: Name or service not known  
[root@ip-172-31-35-172 ~]# ping load_balancer -c4  
PING load_balancer (100.31.244.11) 56(84) bytes of data.  
  
--- load_balancer ping statistics ---  
4 packets transmitted, 0 received, 100% packet loss, time 3126ms  
[root@ip-172-31-35-172 ~]# |
```



It works!



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to fs.com/nginx.

Thank you for using nginx.

```
# utilize system-wide crypto-policies
ssl-default-bind-ciphers PROFILE=SYSTEM
ssl-default-server-ciphers PROFILE=SYSTEM

#-----
# common defaults that all the 'listen' and 'backend' sections will
# use if not designated in their block
#-----
defaults
    mode                http
    log                 global
    option              httplog
    option              dontlognull
    option http-server-close
    option forwardfor   except 127.0.0.0/8
    option               redispatch
    retries             3
    timeout http-request 10s
    timeout queue       1m
    timeout connect     10s
    timeout client      1m
    timeout server      1m
    timeout http-keep-alive 10s
    timeout check       10s
    maxconn             3000

#-----
# main frontend which proxys to the backends
#-----
frontend main
    bind *:80
    acl url_static      path_beg       -i /static /images /javascript /stylesheets
    acl url_static      path_end       -i .jpg .gif .png .css .js

    use_backend static   if url_static
    default_backend      app

#-----
# static backend for serving up images, stylesheets and such
#-----
backend static
    balance roundrobin
    server static 127.0.0.1:4331 check

#-----
# round robin balancing between the various backends
#-----
backend app
    balance roundrobin
    server app1 127.0.0.1:5001 check
    server app2 127.0.0.1:5002 check
    server app3 127.0.0.1:5003 check
    server app4 127.0.0.1:5004 check
    server app5 32.197.239.200:80 check
    server app6 34.238.162.45:80 check
-- INSERT --
```



```
root@ip-172-31-32-181:~  
[root@ip-172-31-32-181 ~]# vi /etc/haproxy/haproxy.cfg  
[root@ip-172-31-32-181 ~]# systemctl enable haproxy  
Created symlink /etc/systemd/system/multi-user.target.wants/haproxy.service → /usr/lib/systemd/system/haproxy.service.  
[root@ip-172-31-32-181 ~]# systemctl start haproxy  
[root@ip-172-31-32-181 ~]# systemctl status haproxy  
● haproxy.service - HAProxy Load Balancer  
   Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: disabled)  
   Active: active (running) since Tue 2026-06-30 17:43:36 UTC; 11s ago  
     Process: 27264 ExecStartPre=/usr/sbin/haproxy -f $CONFIG -f $CFGDIR -c -q $OPTIONS (code=exited, s>  
    Main PID: 27274 (haproxy)  
     Status: "Ready."  
       Tasks: 3 (limit: 1059)  
      Memory: 7.9M  
         CPU: 54ms  
    CGroup: /system.slice/haproxy.service  
            └─27274 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/>  
              27281 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/>  
  
Jun 30 17:43:36 ip-172-31-32-181.ec2.internal haproxy[27274]: [ALERT] (27274) : config : parsing [/>  
Jun 30 17:43:36 ip-172-31-32-181.ec2.internal haproxy[27274]: [NOTICE] (27274) : New worker (27281) >  
Jun 30 17:43:36 ip-172-31-32-181.ec2.internal haproxy[27274]: [NOTICE] (27274) : Loading success.>  
Jun 30 17:43:36 ip-172-31-32-181.ec2.internal systemd[1]: Started haproxy.service - HAProxy Load Balan>  
Jun 30 17:43:36 ip-172-31-32-181.ec2.internal haproxy[27281]: [WARNING] (27281) : Server static/stati>  
Jun 30 17:43:36 ip-172-31-32-181.ec2.internal haproxy[27281]: [ALERT] (27281) : backend 'static' has>  
Jun 30 17:43:36 ip-172-31-32-181.ec2.internal haproxy[27281]: [WARNING] (27281) : Server app/app1 is >  
Jun 30 17:43:37 ip-172-31-32-181.ec2.internal haproxy[27281]: [WARNING] (27281) : Server app/app2 is >  
Jun 30 17:43:37 ip-172-31-32-181.ec2.internal haproxy[27281]: [WARNING] (27281) : Server app/app3 is >  
Jun 30 17:43:37 ip-172-31-32-181.ec2.internal haproxy[27281]: [WARNING] (27281) : Server app/app4 is >  
lines 1-23/23 (END)
```

← → ↻ ⚠ Not secure 100.31.244.11

It works!